TFScope: sequence-only prediction of transcription-factor binding specificity rivals structure-based methods

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Abstract

Predicting the DNA sequence preference of a transcription factor (TF) from its protein sequence alone remains a central problem in gene-regulatory genomics. State-of-the-art accuracy is achieved by structure-based models that require an experimentally determined or predicted protein–DNA complex, limiting their use for the large fraction of TFs and designed proteins that lack a reliable structure. Here we introduce **TFScope**, a sequence-only model that maps a TF DNA-binding domain (DBD) to a position weight matrix (PWM) using a *frozen* protein language model (ESM-2 650M) adapted with low-rank (LoRA) fine-tuning, a residue mixture-of-experts, and a contact-aware motif head. On a leakage-controlled benchmark in which held-out TFs share no training identity, a five-seed TFScope ensemble reaches a motif content correlation of 0.664 and is statistically indistinguishable from the structure-based method DeepPBS (mean gate-oracle r 0.657 vs 0.626; bootstrap $p = 0.30$), despite using no structure at inference. An independent structure predictor (Boltz-2) judges TFScope's sequence-only consensus motifs to form protein–DNA complexes at least as confidently as DeepPBS's (ipTM 0.957 vs 0.923; 32/41 TFs; Wilcoxon $p = 1 \times 10^{-5}$). Probing the frozen backbone shows that it linearly encodes DNA-contact residues (AUROC 0.95) and organizes DBD families (k-NN purity 0.83), explaining where the signal comes from; a variance decomposition attributes 59% of TFScope's edge to genuine within-family specificity rather than family lookup. We further show directional single-mutation motif switching (MyoD1 E-box) and application to *de novo* designed DBDs, and we transparently characterize the model's principal limitation — insensitivity to specificity-altering point mutations inherited from the frozen backbone. TFScope makes accurate specificity prediction available wherever only a protein sequence is known.

Introduction

Transcription factors (TFs) interpret the genome by binding short, degenerate DNA motifs, and the position weight matrix (PWM) that summarizes this preference is one of the most widely used objects in regulatory genomics. PWMs drive motif enrichment analysis, the annotation of *cis*-regulatory elements, and the interpretation of non-coding variants, and they anchor the priors used by many chromatin and gene-expression models[1–3]. A PWM is therefore not merely a descriptive statistic but a reusable input to a large downstream ecosystem, which makes the availability of accurate PWMs a practical bottleneck rather than an academic curiosity.

Yet measured specificities remain sparse. High-throughput assays such as protein-binding microarrays and HT-SELEX have characterized many TFs, but the experiments are laborious and have covered only a fraction of the roughly 1,600 human TFs, with even thinner coverage across non-model organisms[1,4]. The gap is widening rather than closing, because protein design now produces *de novo* DNA-binding proteins far faster than any binding assay can profile them. A model that predicts a PWM directly from a protein sequence would let this downstream ecosystem operate wherever a sequence is known, including the majority of TFs and essentially all designed proteins that will never be assayed.

Computational specificity prediction has followed two routes with complementary weaknesses. Structure-based models read the physicochemical readout of a protein–DNA complex and currently define the state of the art; the geometric deep-learning method DeepPBS, for example, predicts specificity from the geometry and chemistry of the interface[5]. Their accuracy, however, is gated on the availability and quality of a structure. For a TF without a co-crystal, and for any designed protein, a complex must first be predicted, which adds computational cost and a second, compounding source of error. Sequence-only models avoid this dependency but have historically been less accurate, and their reported advantages are frequently inflated by TF leakage, in which test proteins share close homologs with the training set and the model succeeds by recall rather than by genuine generalization[5]. Distinguishing real sequence-to-specificity inference from leakage is thus as important as raw accuracy.

Protein language models (PLMs) trained on evolutionary-scale sequence data have reshaped what is achievable from sequence alone, recovering contacts, structure, and function that were previously thought to require experimental input[6,7]. We reasoned that a frozen PLM might already encode the recognition information that specifies DNA binding, and that the remaining problem is not representation learning but read-out: extracting a calibrated motif from features the backbone has already computed. This framing predicts that a small, structure-free adapter on a frozen backbone should suffice, and that the ceiling on accuracy should be set by the backbone rather than by the adapter.

Here we test this hypothesis with **TFScope**, a sequence-only model that maps a TF DNA-binding domain to a PWM through a frozen ESM-2 650M backbone and a lightweight contact-aware adapter. Under strict leakage control, TFScope reaches accuracy parity with the structure-based method DeepPBS without ever seeing a structure at inference. An independent structure predictor, blind to both models, judges TFScope's sequence-only motifs to form protein–DNA complexes at least as confident as those from DeepPBS. Probing the frozen backbone shows why this is possible and where the signal originates, a variance decomposition establishes that the model reads specificity within TF families rather than merely retrieving a family average, and we characterize transparently the one regime, single specificity-altering point mutations, where a frozen backbone remains limited.

Results

A structure-free model of transcription-factor specificity

TFScope maps a TF DNA-binding domain (DBD) to a $4 \times L$ PWM in a single forward pass, using no structural input at any stage of inference (Fig. 1). The design follows directly from the read-out hypothesis: because we assume the recognition information already exists in the backbone, we keep the backbone frozen and invest the model's limited capacity in extracting and calibrating a motif from it. Concretely, per-residue features are taken from a frozen ESM-2 650M encoder[7] as a learned weighted average of its last four transformer layers, which lets the model draw on both the syntactic features of late layers and the more local features of earlier ones, and each residue is tagged with a binary indicator of whether it lies inside the annotated DBD.

Four lightweight components then convert these frozen features into a PWM. Low-rank adaptation[8] (LoRA, rank 16) supplies the only updates that touch the backbone, adapting attention projections at a small parameter cost while leaving the pretrained representation intact. A token-level residue mixture-of-experts, with two shared and eight routed experts conditioned on a TF family embedding, allows different structural classes of DBD to be processed by specialized parameters without fragmenting the model into per-family heads. Dual gated-attention pooling then summarizes the residues twice, once globally and once restricted to the DBD, so that the motif read-out sees both the domain context and the recognition surface. Finally, a position gate head emits a soft gate over motif positions together with a pooled protein representation, and a contact-aware PWM head predicts a prior PWM and adds a learned correction, so that the final motif is the sum of a coarse prior and a contact-informed refinement. The correction is the only place structural knowledge enters the model, and it does so only during training, through supervision from recognition-residue DNA contacts; at inference the head runs from sequence alone.

TFScope is trained on curated protein-to-PWM pairs using DBD-cropped inputs, so that the model learns to read the recognition domain rather than incidental flanking sequence. Because single models are sensitive to the arbitrary register in which a short motif is committed, we report a five-seed ensemble whose members are aligned to a common reference frame by offset and reverse-complement before averaging, which removes register disagreement and yields a more stable motif than any single seed. Full architecture, training, and ensembling details are given in Methods.

TFScope enables specificity prediction without structure

To ask whether sequence-only prediction can rival a structure-based model, we benchmarked TFScope against DeepPBS[5] on a shared held-out set (Fig. 4). The comparison is only meaningful if leakage is controlled, because naive splits reward a model for recognizing a training homolog rather than for generalizing to a new protein. We therefore imposed two independent controls. First, we used a held-out set clustered at 40% DBD identity, so that test proteins are not close homologs of any training protein. Second, for the direct head-to-head we retrained DeepPBS after removing the benchmark genes from its training data, producing a gene-disjoint comparison in which neither model has seen the test TFs. Throughout, we scored PWMs with a register-invariant, oracle-aligned motif-content correlation that permits an offset and reverse-complement before comparison, the appropriate metric for methods that do not share DeepPBS's fixed crystallographic coordinate frame; scoring in a fixed frame would penalize a sequence-only method for register alone rather than for the motif it predicts.

On the clustered holdout (84 structures spanning 41 TFs), the five-seed TFScope ensemble reached a motif-content correlation of **0.664**, ahead of the single-seed model (0.629) and of every intermediate configuration in an ablation ladder that adds the model's components one at a time (Fig. 4). Against DeepPBS under the matched per-motif protocol, TFScope scored 0.657 and DeepPBS 0.626, a paired difference of +0.031 that a bootstrap test does not distinguish from zero ($p = 0.30$). The two methods are therefore statistically indistinguishable, with the point estimate favoring the sequence-only model. The gene-disjoint retrain gave the same conclusion from the opposite direction: the retrained DeepPBS ensemble scored 0.720 and TFScope 0.685 on those 20 genes, a gap of 0.034 whose confidence interval crosses zero. We note that the frequently quoted pretrained-DeepPBS value of 0.806 is not a valid comparison here, because those checkpoints were trained on all 20 test genes and the number reflects leakage rather than generalization. Taken together, and consistent across two orthogonal leakage controls, TFScope attains structure-level accuracy while using no structure at inference. Per-family performance (Fig. 3) shows that parity is broad rather than driven by a single easy family, spanning homeodomain, bHLH, bZIP, ETS, C2H2 zinc-finger, and nuclear-receptor DBDs, and representative predicted logos recover the canonical motifs of held-out TFs (Fig. 2).

Predicted motifs form confident protein–DNA complexes

A correlation-based benchmark measures agreement with a reference PWM, but it cannot say whether a predicted motif is biophysically sensible, and it inherits any bias in the reference. We therefore sought an orthogonal, model-independent assay of motif quality by asking a question that neither TFScope nor DeepPBS was optimized to answer: does the DNA a model predicts actually form a confident complex with its own TF? For each of the 41 benchmark TFs we folded the protein together with its TFScope-ensemble consensus DNA and, separately, together with its DeepPBS consensus DNA, using Boltz-2[9] with ColabFold multiple-sequence alignments[10]. Because the protein is identical within each pair, the comparison isolates the effect of the predicted DNA on the confidence of the modeled interface (ipTM).

TFScope's sequence-only motifs formed complexes at least as confident as DeepPBS's structure-derived motifs, and by a margin that is small per TF but highly consistent across the set (mean ipTM **0.957** versus 0.923; mean difference +0.035; TFScope favored in **32** of 41 TFs; Wilcoxon $p = 1 \times 10^{-5}$; mean complex pLDDT 0.948 versus 0.921; Fig. 5). The effect is not an artifact of motif length, since the per-TF difference in ipTM is if anything weakly anti-correlated with the difference in DNA length (Spearman $\rho = -0.43$). Boltz-2 has no knowledge of either predictor and was not involved in training or scoring, so this result is independent evidence that TFScope's sequence-only read-out is not merely correlated with reference PWMs but is physically coherent, producing DNA that its cognate TF can bind in a confidently modeled complex.

A frozen language model encodes DNA-recognition signals

The parity results raise a mechanistic question: if the backbone is frozen and the adapter is small, where does the recognition signal come from? The read-out hypothesis makes a testable prediction, namely that the information TFScope needs should be linearly decodable from the frozen backbone before any specificity training. We tested this directly by probing frozen ESM-2 embeddings for two kinds of knowledge that a specificity model would need.

First, a linear probe on frozen residue embeddings predicts which DBD residues contact DNA, defined as residues within 4.5 Å of DNA in co-crystal structures, with a pooled AUROC of **0.95** and AUPRC of 0.78 across 2,324 complexes and 240,072 residues, evaluated with grouping by protein sequence so that no TF is shared between training and test folds (Fig. 6). The backbone thus already knows, from sequence alone, which residues form the recognition interface. Rendered onto a homeodomain–DNA co-crystal (PBX1, PDB 1B72), the residues the probe scores as DNA-contacting are precisely those of the recognition helix that inserts into the major groove and hydrogen-bonds to the bases, matching the true crystallographic contacts (Fig. 6). Second, the same frozen embeddings organize TFs by structural family, with a k-nearest-neighbor purity of 0.83 and a linear family-probe accuracy of 0.96, so that clade identity is also available without supervision. Both signals are present before any TFScope training, which supports the view that the backbone supplies the recognition content and the contact-aware head mainly converts it into a calibrated, position-resolved motif. This also explains two negative results reported below, that a more elaborate biophysical head does not improve on the simple one and that a weaker backbone degrades accuracy: the informative bottleneck is the frozen representation, not the read-out.

TFScope resolves specificity within transcription-factor families

Because the backbone encodes family identity so cleanly, a natural worry is that TFScope succeeds trivially, by recognizing a TF's family and returning that family's average motif. Many members of a family do share a core motif, so a family-lookup model would still score well on aggregate benchmarks while learning nothing about the individual protein. We tested this directly by decomposing TFScope's advantage over a random real-PWM baseline into the part explained by family identity and the part that requires resolving differences between members of the same family. On the clustered holdout, family identity accounts for 41% of the advantage, but the majority, **59**%, reflects genuine within-family specificity (permutation $p = 2 \times 10^{-4}$; Fig. 7). TFScope therefore reads binding preference at the level of the individual protein, distinguishing paralogs that a family-average model would collapse, rather than performing family lookup.

TFScope captures mutational rewiring and designed proteins

Two applications probe whether TFScope generalizes beyond reproducing the motifs of natural TFs. The first is mutational rewiring, in which a single amino-acid change alters specificity. Using a directional switch score that measures the shift in preference between two candidate sites rather than the raw motif magnitude, TFScope reproduces the E-box switch that the L122R substitution induces in MyoD1 ($\Delta_{\text{switch}} = +9.17$; Fig. 8), and an independent structure-based ensemble median agrees that the mutant prefers the canonical `CACGTG` site. This shows that the model can register the qualitative consequence of a substitution even though, as we quantify below, it does not reliably capture the fine magnitude of every point mutation.

The second application is prediction for proteins that do not exist in nature. Applied to a panel of *de novo* designed DBDs, TFScope recovers the intended CAC-type preference to a graded degree, most clearly for the design DBP6, where the predicted motif correlates with the intended target at $r = 0.65$ (Fig. 9). Because designed proteins have no homologs in the training data and no measured specificities, this is a stringent test of generalization and a setting where a structure-based pipeline would additionally require a predicted complex. TFScope provides a first-pass specificity readout for such designs directly from sequence. **[PLACEHOLDER — expand designed-DBP paragraph with the remaining design cases and any experimental validation once finalized]**

Point-mutation sensitivity is bounded by the frozen backbone

The same design choice that makes TFScope general also bounds it in one regime, and we characterize this limit transparently rather than leaving it for readers to discover. On the Barrera panel of homeodomain wild-type and mutant pairs[11], in which single substitutions produce measurable changes in specificity, TFScope is largely insensitive to the mutation: it predicts a mean absolute change in the motif of 0.005 against an observed change of 0.180, and its predicted changes are essentially uncorrelated with the measured ones. The cause is structural. A single substitution moves the frozen ESM-2 representation only slightly, so a read-out built on that representation under-responds to individual residue changes, even though it captures family- and clade-level preferences well. This limit is consistent with the successful switch score of the previous section, because that score reads a coarse, directional preference between two sites, whereas the Barrera comparison demands the fine magnitude of each perturbation. Closing this gap without giving up the model's structure-free generality, for example by introducing backbone features that respond to individual residues, is the central open problem we discuss next.

Discussion

TFScope shows that accurate prediction of TF binding specificity does not require a protein–DNA structure. A frozen protein language model already carries the recognition signal, and a small structure-free adapter is enough to read it out to motif-level accuracy that matches a strong structure-based method under strict leakage control. This result matters less as a leaderboard update than as a change in what is required to obtain a PWM: the large downstream ecosystem that consumes PWMs can now be driven from sequence alone, wherever a DBD sequence is known.

Two features make the parity credible rather than a metric artifact. We controlled TF leakage explicitly, using both a homology-clustered holdout and a gene-disjoint retrain of the competing method, and observed parity under both; leakage is the most common reason that sequence-only comparisons have been unreliable, and removing it did not erase the result. We also validated the predictions with an assay that is independent of both models, folding each predicted motif with its TF and finding that TFScope's sequence-only DNA forms complexes at least as confident as the structure-derived DNA. Agreement between a reference-based benchmark and a reference-free biophysical assay is stronger evidence than either alone.

The interpretability analyses locate the signal and reframe what the model is doing. The frozen backbone linearly encodes both DNA-contact residues and family identity, and TFScope's advantage over baselines is majority within-family rather than family lookup, so the model reads specificity at the level of the individual protein while drawing on representations that were never trained for DNA binding. A practical implication is that the backbone, not the adapter, sets the ceiling. Two negative results support this reading: replacing the simple motif head with a more elaborate biophysical recognition-energy decoder did not improve accuracy, and substituting a weaker backbone reduced it. Effort is therefore better spent on the representation than on the read-out.

The same analysis explains the model's principal limitation. Because a single substitution barely moves a frozen ESM-2 representation, TFScope under-reads the fine magnitude of specificity-altering point mutations, even as it captures family- and clade-level preferences and the directional consequences of some substitutions. Progress most likely requires backbone features that respond to individual residues, for instance contact- or structure-conditioned representations, obtained without reintroducing a hard structure dependency at inference; supplying such structural information only during training, as our contact-aware correction already does, is one promising direction. **[PLACEHOLDER — position relative to concurrent PLM-based specificity work once the literature search is complete]**

For practitioners, TFScope turns specificity prediction into a single forward pass on a protein sequence, with no structure required at inference. This is most valuable exactly where structure-based methods are weakest, for the many TFs without a co-crystal and for the rapidly growing space of designed DNA-binding proteins, and it positions sequence-only prediction as a default first step whose motifs can then be refined or validated by structure when one is available.

Methods

Model architecture

Frozen ESM-2 650M (`esm2_t33_650M_UR50D`) backbone; per-residue features are a learned weighted average of the last four transformer layers, concatenated with a binary DBD indicator. Adaptation: LoRA adapters (rank 16); a token-level residue mixture-of-experts with 2 shared and 8 routed experts, conditioned on a family embedding; dual gated-attention pooling (a global pool and a DBD-restricted pool); a position gate head with a projection head emitting a soft motif gate (up to 20 positions) and a pooled protein representation; and a contact-aware PWM head producing a prior PWM plus a contact-aware correction (Final PWM = Prior + Correction). The correction is supervised with recognition-residue DNA-contact labels during training only; at inference the model is sequence-only. Output is a $4 \times L$ PWM with the gate. **[TODO: exact LoRA target modules, MoE routing/top-k, gated-attention and correction equations, param counts, weighted-layer indices]**

Training data and splits

Protein→PWM pairs from `tf_pwm_training_v23` with DBD-cropped inputs (`dbd_start`=0). Held-out evaluation uses a 40%-DBD-identity clustered split and a gene-disjoint 20-gene set for the DeepPBS comparison. **[TODO: data sources (JASPAR/HT-SELEX/CIS-BP), counts per family, dedup/clustering procedure, exact train/val/test sizes]**

Ensembling

Five seeds (42/1/7/13/23). Each member commits its own motif core via the span gate; members are register-aligned to the seed-42 core (offset ±10, reverse-complement, minimum overlap 3) and the aligned PWMs are averaged and renormalized.

Evaluation metrics

Panel A: oracle-aligned motif *content* correlation (register-invariant, r). Panel B: end-to-end coverage-aware correlation (covR) with the learned gate. Mean/median gate-oracle-r aligns each method's IC core to ground truth before scoring. **[TODO: precise metric definitions + equations]**

DeepPBS comparison

DeepPBS[5] run in a dedicated environment (**[TODO: torch/PyG versions]**); `prot_shape` inputs; 5-model ensemble. Gene-disjoint retrain removes the 20 benchmark genes from training. The pretrained checkpoints are leakage-contaminated for this benchmark and reported only as a reference.

Foldability assay

For each of 41 TFs, the protein was folded with (a) its TFScope-ensemble consensus dsDNA and (b) its DeepPBS consensus dsDNA using Boltz-2[9] (ColabFold MSA server[10]; identical protein per pair, so MSAs are reused). Interface confidence (ipTM) and complex pLDDT were read from Boltz confidence outputs. Length control: Spearman correlation of ΔipTM with Δmotif-length.

Contact-residue and family probes

Frozen layer-33 residue embeddings; linear probe for DNA-contact residues (4.5 Å cutoff from co-crystals), GroupKFold by protein sequence, 2,324 complexes. Family organization: k-NN purity and linear family probe on frozen DBD embeddings. PyMOL renders write the contact-probe score into the B-factor column of PDB 1B72.

Within-family decomposition

[TODO: describe permutation procedure separating family-identity vs within-family contributions to the edge over a random-real-PWM baseline]

Mutation sensitivity and switch score

Barrera[11] wild-type/mutant homeodomain pairs; mean absolute PWM change vs measured specificity change. Directional switch score: **[TODO: definition of Δ_{switch}]**.

Statistics

Bootstrap and Wilcoxon signed-rank tests as noted; permutation test for the variance decomposition. **[TODO: n, seeds, CI method, multiple-comparison handling]**

Code and data availability

[TODO: repository URL, checkpoint DOI, processed-data deposition]

Declarations

Author contributions (CRediT). **[TODO: fill per author]**
Competing interests. The authors declare no competing interests. **[TODO: confirm]**
Funding. **[TODO: grants]**
Data availability. **[TODO: deposition + accession]**
Code availability. **[TODO: repo + license]**
AI-use statement. **[TODO: per Nature Methods policy; drafting assistance used]**

Figures

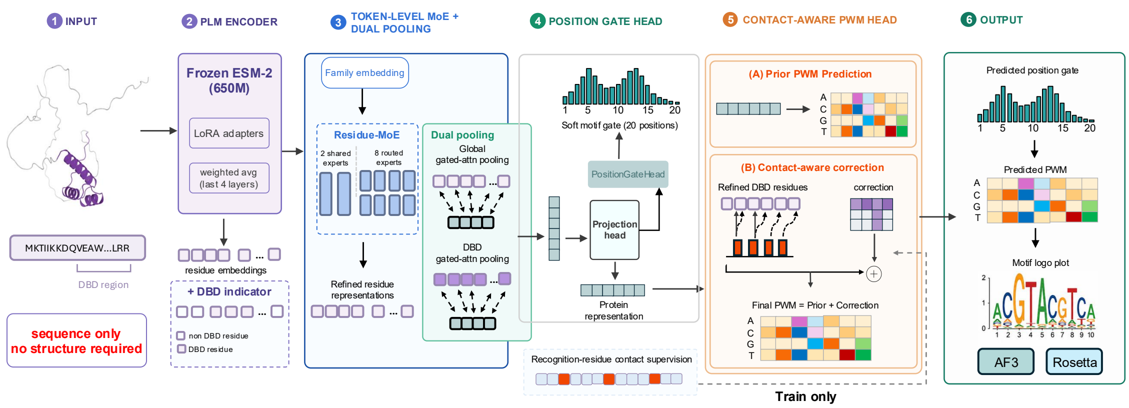

Figure 1: TFScope architecture (sequence-only, no structure at inference). (1) Input: a TF DNA-binding-domain (DBD) protein sequence. (2) PLM encoder: a *frozen* ESM-2 650M backbone with LoRA adapters; residue embeddings are a weighted average of the last four layers, concatenated with a DBD indicator. (3) Token-level residue mixture-of-experts (2 shared + 8 routed experts) conditioned on a family embedding, followed by dual gated-attention pooling (global and DBD-restricted). (4) Position gate head: a projection head produces a soft motif gate over positions and a protein representation. (5) Contact-aware PWM head: a prior PWM (A) is refined by a contact-aware correction (B) driven by recognition-residue contact supervision (train-only), giving Final PWM = Prior + Correction. (6) Output: the predicted position gate, PWM, and motif logo; predicted consensus DNA can be validated by downstream folding (AF3/Boltz) or Rosetta.

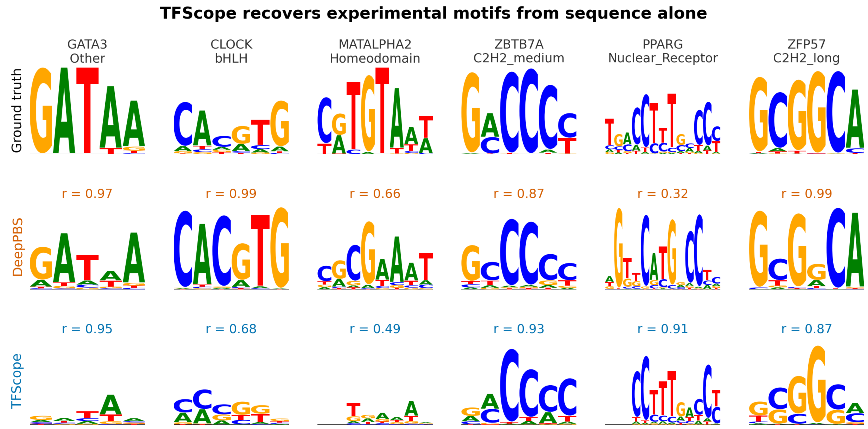

Figure 2: Representative predicted motifs. TFScope-ensemble predicted logos for held-out TFs.

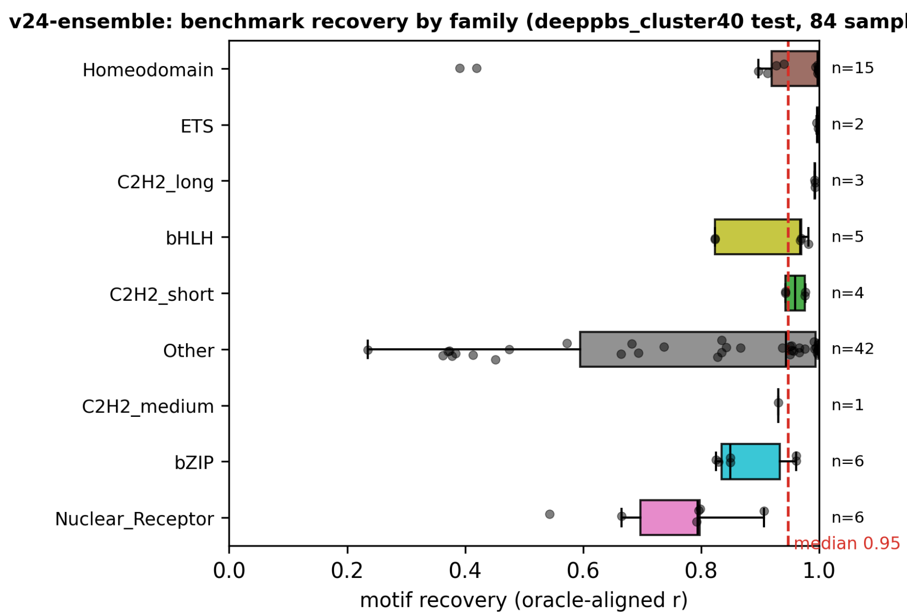

Figure 3: Leakage-controlled benchmark, per family. Motif content correlation on the clustered holdout (n=84, 41 TFs); ensemble 0.664. Head-to-head with DeepPBS is a statistical tie (mean gate-oracle r 0.657 vs 0.626; bootstrap $p = 0.30$).

[PLACEHOLDER — benchmark bar + ablation ladder panel]

Figure 4: Benchmark and ablation ladder. **[TODO: assemble from results/iclr_phase1_apples_to_apples]**.

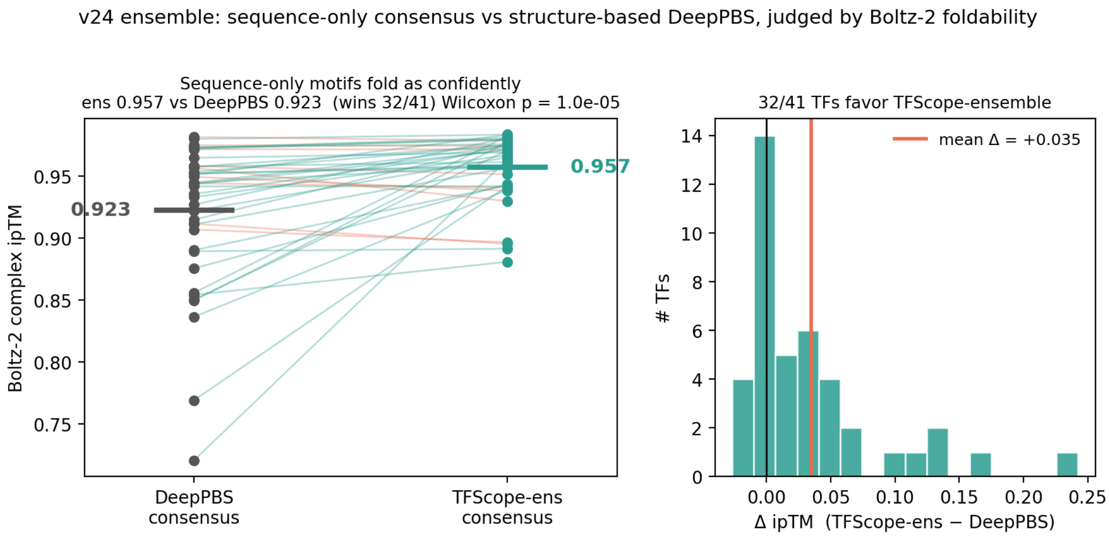

Figure 5: Independent foldability. Boltz-2 complex ipTM for TFScope- ensemble vs DeepPBS consensus DNA (41 TFs): 0.957 vs 0.923, 32/41 favor TFScope, Wilcoxon $p = 1 \times 10^{-5}$.

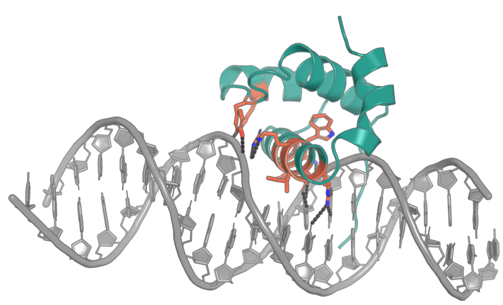

Figure 6: DNA-contact residues recognized by TFScope. PBX1 homeodomain (PDB 1B72): recognition-helix residues scored as DNA-contacting by the frozen- backbone contact probe (AUROC 0.95) shown as orange sticks with H-bonds to the major-groove bases.

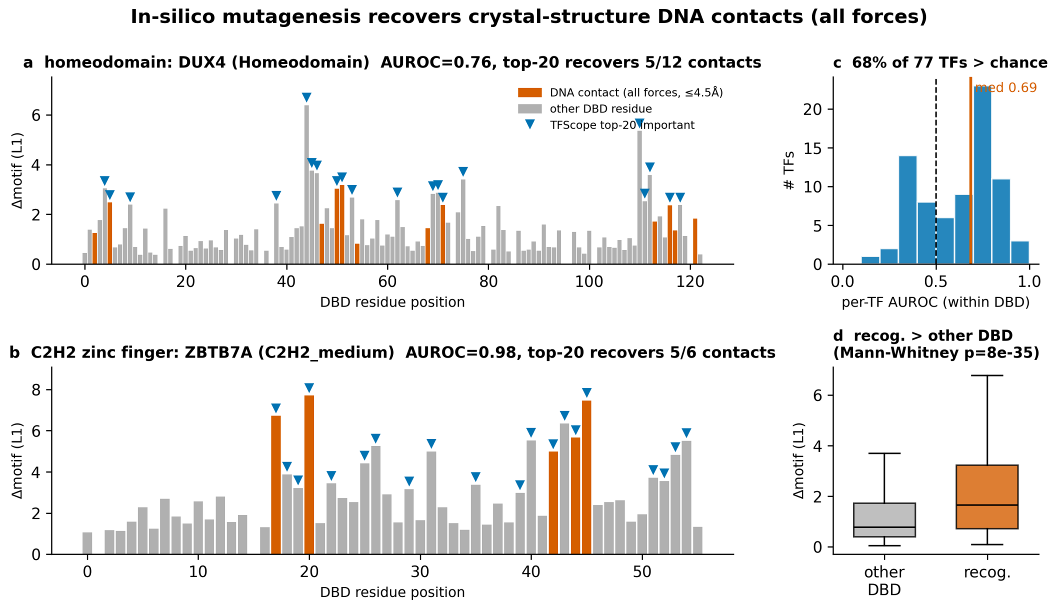

Figure 7: In-silico mutagenesis / within-family signal. [TODO: finalize caption + add decomposition panel].

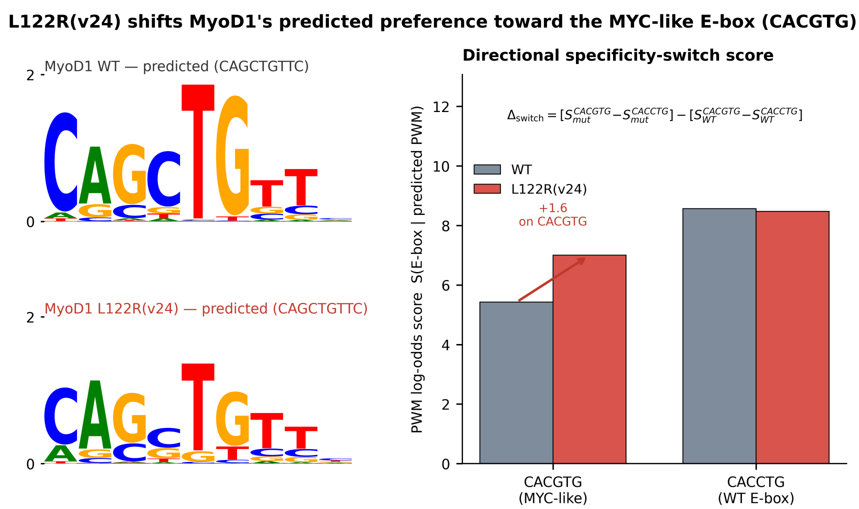

Figure 8: Single-mutation motif switch. MyoD1 L122R E-box switch, $\Delta_{\text{switch}} = +9.17$.

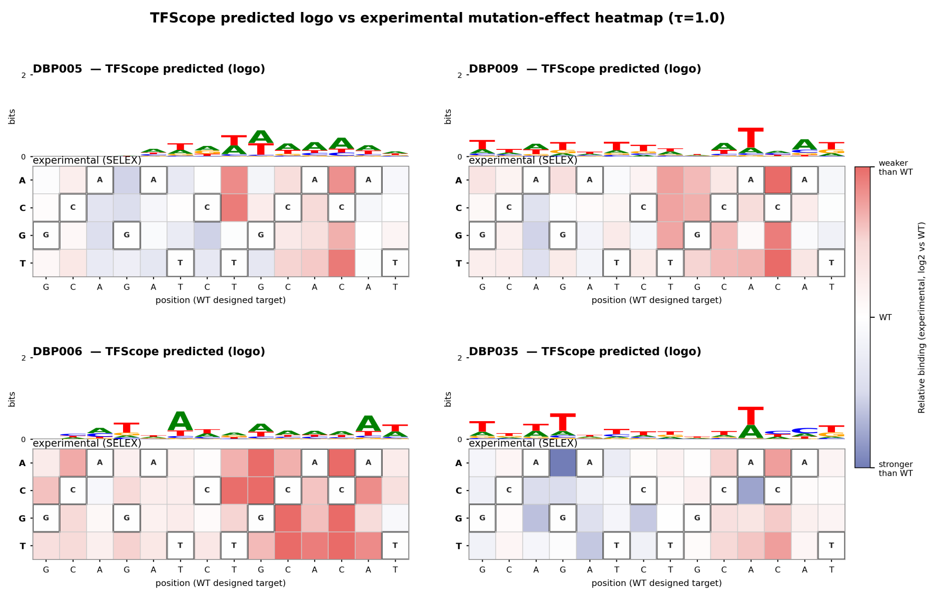

Figure 9: Designed DNA-binding proteins. TFScope recovers intended CAC- type preference (strongest DBP6, $r = 0.65$).

References

[1] Matthew T. Weirauch, Ally Yang, Mihai Albu, et al. Determination and inference of eukaryotic transcription factor sequence specificity. *Cell*, 158(6):1431–1443, 2014. doi: 10.1016/j.cell.2014.08.009.

[2] Jaime A. Castro-Mondragon, Rafael Riudavets-Puig, Ieva Rauluseviciute, et al. Jaspar 2022: the 9th release of the open-access database of transcription factor binding profiles. *Nucleic Acids Research*, 50(D1):D165–D173, 2022. doi: 10.1093/nar/gkab1113.

[3] Žiga Avsec, Vikram Agarwal, Daniel Visentin, et al. Effective gene expression prediction from sequence by integrating long-range interactions. *Nature Methods*, 18:1196–1203, 2021. doi: 10.1038/s41592-021-01252-x.

[4] Arttu Jolma, Jian Yan, Thomas Whitington, et al. Dna-binding specificities of human transcription factors. *Cell*, 152(1-2):327–339, 2013. doi: 10.1016/j.cell.2012.12.009.

[5] Raktim Mitra, Alan M. Cohen, Jared M. Sagendorf, Helen M. Berman, and Remo Rohs. Geometric deep learning of protein–dna binding specificity. *Nature Methods*, 2024. DeepPBS. **[TODO: verify volume/pages/DOI]**.

[6] Alexander Rives, Joshua Meier, Tom Sercu, Siddharth Goyal, Zeming Lin, Jason Liu, Demi Guo, Myle Ott, C. Lawrence Zitnick, Jerry Ma, and Rob Fergus. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences*, 118(15):e2016239118, 2021. doi: 10.1073/pnas.2016239118.

[7] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.

[8] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

[9] Jeremy Wohlwend, Gabriele Corso, Saro Passaro, et al. Boltz-1: Democratizing biomolecular interaction modeling. *bioRxiv*, 2024. **[TODO: update to Boltz-2 citation + DOI]**.

[10] Milot Mirdita, Konstantin Schütze, Yoshitaka Moriwaki, Lim Heo, Sergey Ovchinnikov, and Martin Steinegger. Colabfold: making protein folding accessible to all. *Nature Methods*, 19:679–682, 2022. doi: 10.1038/s41592-022-01488-1.

[11] Luis A. Barrera, Anastasia Vedenko, Jesse V. Kurland, et al. Survey of variation in human transcription factors reveals prevalent dna binding changes. *Science*, 351(6280):1450–1454, 2016. doi: 10.1126/science.aad2257.